1. What is the basic building block of every artificial neural network?

A) Matrix
B) Artificial Neuron
C) Bias
D) Kernel
Answer: B) Artificial Neuron

2. Which activation function is most commonly used in the perceptron model?

A) Sigmoid
B) Step function
C) ReLU
D) Tanh
Answer: B) Step function

3. A perceptron can solve which type of problems?

A) Non-linear separable
B) Only linear separable
C) XOR
D) Clustering
Answer: B) Only linear separable

4. What is the main purpose of using activation functions in neural networks?

A) To speed up training
B) To introduce non-linearity
C) To reduce overfitting
D) To increase memory
Answer: B) To introduce non-linearity

5. A multilayer perceptron (MLP) must have at least which three layers?

A) Two input layers, one output layer
B) One input, one hidden, one output layer
C) Two output layers, one input layer
D) Three hidden layers
Answer: B) One input, one hidden, one output layer

6. Which of the following is NOT an activation function commonly used in neural networks?

A) Sigmoid
B) ReLU
C) Softmax
D) Fourier
Answer: D) Fourier

7. What is the primary purpose of the loss function in neural networks?

A) To increase computation speed

B) To evaluate model performance
C) To regularize the weights
D) To initialize parameters
Answer: B) To evaluate model performance

8. What is backpropagation used for in neural networks?

A) Initializing weights
B) Optimizing the activation function
C) Updating the weights during training
D) Decreasing bias
Answer: C) Updating the weights during training

9. Which challenge is most associated with training deep neural networks?

A) Overfitting
B) Underfitting
C) Vanishing gradient
D) All of the above
Answer: D) All of the above

10. In neural network training, which term refers to the model being too simple to capture the data patterns?

A) Overfitting
B) Vanishing gradient
C) Underfitting
D) Batch normalization
Answer: C) Underfitting

11. Which of the following is the main purpose of an activation function in a neural network?

a) Reduce dimensionality
b) Introduce non-linearity
c) Perform normalization
d) Optimize learning rate
Answer: b) Introduce non-linearity Explanation: Activation functions allow neural networks to learn complex, non-linear patterns. Without them, the network is just a linear model.

12. Which of the following activation functions can cause the vanishing gradient problem?

a) ReLU
b) Sigmoid
c) Tanh
d) Both (b) and (c)
Answer: d) Both (b) and (c) Explanation: Sigmoid and tanh squash inputs to small ranges, leading to very small gradients for large inputs, causing vanishing gradients.

13. What is the output range of the Sigmoid activation function?

a) [-1, 1]
b) [0, 1]
c) (-∞, ∞)
d) [0, ∞)

Answer: b) [0, 1] Explanation: Sigmoid maps values into the range [0, 1], making it useful for Probabilities.

14. What is the main drawback of the ReLU activation function?

a) High computation cost
b) Exploding gradients
c) Dying ReLU problem
d) Non-differentiability at zero

Answer: c) Dying ReLU problem Explanation: For negative inputs, ReLU outputs zero, which may cause some neurons to "die" and stop learning.

15. The Tanh activation function is preferred over Sigmoid in some cases because:

a) It is computationally faster
b) Its range is [-1, 1]
c) It avoids exploding gradients
d) It is linear for large values

Answer: b) Its range is [-1, 1] Explanation: Tanh is zero-centered, unlike sigmoid, which helps optimization in many cases.

16. Which of the following is a smooth approximation of ReLU that allows small gradients for negative inputs?

a) Leaky ReLU
b) Sigmoid
c) Softmax
d) Step function

Answer: a) Leaky ReLU Explanation: Leaky ReLU allows a small slope (e.g., 0.01x) for negative
inputs, avoiding dead neurons.

17. Softmax activation is mainly used in which type of neural network layer?

a) Hidden layers
b) Input layers
c) Output layer for multi-class classification
d) Output layer for regression

Answer: c) Output layer for multi-class classification Explanation: Softmax converts logits into probabilities that sum to 1, making it ideal for classification.

18. Which activation function is defined as f(x) = max(0, x)?

a) Sigmoid
b) ReLU
c) Tanh
d) Softplus

Answer: b) ReLU Explanation: ReLU outputs input if positive, else 0.

19. Which of the following activation functions is most likely to cause exploding gradient problems?

a) Sigmoid
b) ReLU
c) Tanh
d) None of the above

Answer: d) None of the above Explanation: Exploding gradients are more related to poor weight initialization or deep networks, not specifically activation functions. However, sigmoid/tanh usually cause vanishing, not exploding gradients.

20. Which activation function is often used in hidden layers of deep CNNs due to efficiency and effectiveness?

a) Sigmoid
b) ReLU
c) Tanh
d) Softmax

Answer: b) ReLU Explanation: ReLU is computationally efficient and reduces vanishing gradients, making it popular in deep learning.

21. **What is the main purpose of backpropagation in neural networks?**
    a) To initialize weights randomly
    b) To propagate inputs forward through the network
    c) To compute gradients of the loss function with respect to weights
    d) To increase the number of hidden layers

✅ **Answer:** c) To compute gradients of the loss function with respect to weights

22. **Which optimization algorithm is most commonly used with backpropagation?**
    a) Breadth-first search
    b) Gradient descent
    c) Reinforcement learning
    d) K-means clustering

✅ **Answer:** b) Gradient descent

23. **In backpropagation, the chain rule of calculus is used to:**
    a) Reduce the number of parameters

b) Compute partial derivatives across multiple layers
c) Normalize input data
d) Prevent overfitting

✅ **Answer:** b) Compute partial derivatives across multiple layers

24. **Which of the following is a common issue in training deep neural networks?**
    a) Gradient vanishing and explosion
    b) Lack of bias terms
    c) Too many output layers
    d) Sparse activation functions only

✅ **Answer:** a) Gradient vanishing and explosion

25. **What does the vanishing gradient problem lead to?**
    a) Weights growing very large
    b) Weights failing to update effectively in early layers
    c) Overfitting to training data
    d) Faster convergence

✅ **Answer:** b) Weights failing to update effectively in early layers

26. **Which activation function is most prone to vanishing gradient problems?**
    a) ReLU
    b) Sigmoid
    c) Tanh
    d) Softmax

✅ **Answer:** b) Sigmoid

27. **The exploding gradient problem typically results in:**
    a) Extremely large weight updates
    b) Very slow convergence
    c) No change in parameters
    d) Decreasing loss without accuracy improvement

✅ **Answer:** a) Extremely large weight updates

28. **Which of the following techniques helps to address vanishing/exploding gradient problems?**
    a) Weight initialization methods (e.g., Xavier, He)
    b) Dropout layers

c) Adding noise to inputs
d) Feature scaling

✅ **Answer:** a) Weight initialization methods (e.g., Xavier, He)

29. **What role does the learning rate play in backpropagation?**
    a) It determines the complexity of the network
    b) It controls the size of weight updates during training
    c) It reduces the number of epochs needed
    d) It prevents gradient vanishing automatically

✅ **Answer:** b) It controls the size of weight updates during training

30. **Overfitting in neural network training can be reduced by:**
    a) Increasing the number of hidden layers
    b) Decreasing the size of the training set
    c) Using regularization techniques such as dropout
    d) Removing activation functions

✅ **Answer:** c) Using regularization techniques such as dropout

31. Overfitting occurs when:

a) The model fits unseen data perfectly
b)The model memorizes training data and fails to generalize
c) The model ignores important features of data
d) The model underestimates patterns in data
Answer: b) The model memorizes training data and fails to generalize

32. High bias usually results in:
a) Overfitting
b) Underfitting
c) High accuracy
d) Low training error
Answer: b) Underfitting

33. High variance usually results in:
a) Stable predictions
b) Overfitting

c) Underfitting
d) Low training error
Answer: b) Overfitting

34. Bias-variance trade-off means:
a) Reducing both bias and variance
b) Reducing bias increases variance and vice versa
c) Bias is independent of variance
d) Only variance matters
Answer: b) Reducing bias increases variance and vice versa

35. Hyperparameters are tuned to:
a) Improve generalization
b) Reduce dataset
c) Eliminate bias
d) Remove variance
Answer: a) Improve generalization

36. Hyperparameters are tuned using:
a) Training set
b) Validation set
c) Test set
d) Noise data
Answer: b) Validation set

37. Validation set is used for:
a) Model training
b) Hyperparameter tuning
c) Final testing
d) Reducing bias
Answer: b) Hyperparameter tuning

38. Usual dataset split is:
a) Training & test only
b) Training, validation & test
c) Training only
d) Validation only
Answer: b) Training, validation & test

39. Early stopping prevents overfitting by:

a) Monitoring training set
b) Monitoring validation set
c) Increasing epochs
d) Reducing dataset
Answer: b) Monitoring validation set

40. When validation error increases but training error decreases, the model is:
a) Well-generalized
b) Underfitting
c) Overfitting
d) Balanced
Answer: c) Overfitting

41. Why is initializing all weights to zero a bad practice in deep neural networks?
- A) It speeds up convergence
- B) It prevents symmetry breaking, leading to identical neurons
- C) It increases gradient values
- D) It improves generalization

Correct Answer: B

42. Which initialization method is recommended for networks using sigmoid/tanh activations?
- A) He Initialization
- B) Xavier Initialization
- C) Zero Initialization
- D) Random Large Weights

Correct Answer: B

43. He Initialization was specifically designed for which type of activation function?
- A) ReLU
- B) Sigmoid
- C) Tanh
- D) Softmax

Correct Answer: A

44. What is the main issue if initial weights are too small in deep networks?

- A) Exploding gradients
- B) Vanishing gradients
- C) Overfitting
- D) Faster convergence

Correct Answer: B

45. The vanishing gradient problem occurs more frequently with which activation functions?
- A) ReLU / Leaky ReLU
- B) Sigmoid / Tanh
- C) Softmax
- D) Maxout

Correct Answer: B

46. Which of the following is a solution to the vanishing gradient problem?
- A) Gradient clipping
- B) Residual Networks (ResNets)
- C) Zero initialization
- D) Increasing learning rate indefinitely

Correct Answer: B

47. Exploding gradients typically cause:
- A) Gradients to shrink to zero
- B) Very large weight updates and unstable training
- C) Slower learning but stable convergence
- D) Model underfitting

Correct Answer: B

48. A common method to control exploding gradients is:
- A) Gradient clipping
- B) Dropout
- C) Batch Normalization
- D) Ensemble methods

Correct Answer: A

49. What is the role of Batch Normalization in training deep models?
- A) Reduces dataset size
- B) Normalizes inputs to each layer, improving gradient flow

- C) Prevents overfitting directly
- D) Eliminates the need for activation functions

Correct Answer: B

50. Which of the following pairs is correct regarding problems and solutions?

- A) Vanishing gradients → Gradient clipping
- B) Exploding gradients → Residual connections
- C) Vanishing gradients → ReLU activation
- D) Exploding gradients → Zero initialization

Correct Answer: C

51. What is the primary problem caused by the vanishing gradient phenomenon?
    A. The gradients become uncontrollably large, causing instability.
    B. The network converges too quickly to a suboptimal solution.
    C. The initial layers of the network stop learning because their weights are not updated effectively.
    D. The learning rate increases automatically, causing oscillations.
    Answer: C. The initial layers of the network stop learning because their weights are not updated effectively.

52. According to the document, which of the following is a recommended solution for the exploding gradient problem? 💣
    A. Using ReLU or leaky ReLU activation functions.
    B. Using Residual Networks (ResNets).
    C. Applying Batch Normalization.
    D. Using gradient clipping.
    Answer: D. Using gradient clipping.

53. Which of the following is listed as a solution specifically for the vanishing gradient problem?
    A. Gradient clipping.
    B. Proper weight initialization.
    C. Using Residual Networks (ResNets).
    D. Using sigmoid/tanh activations.
    Answer: C. Using Residual Networks (ResNets).
54. What is a major disadvantage of standard Gradient Descent (GD) when compared to Stochastic Gradient Descent (SGD)?

A. It has very noisy and erratic convergence.
B. It is computationally expensive and slow for large datasets.
C. It updates weights after looking at only a single data sample.
D. It is less likely to find the true minimum of the loss function.
Answer: B. It is computationally expensive and slow for large datasets.

55. What is the main purpose of adding momentum to Gradient Descent?
A. It adapts the learning rate for each parameter based on past gradients.
B. It helps accelerate convergence, reduce oscillations, and escape local minima.
C. It guarantees that the learning rate will decrease over time.
D. It calculates the gradient on the entire dataset.
Answer: B. It helps accelerate convergence, reduce oscillations, and escape local minima.

56. How does the Nesterov Momentum method fundamentally differ from standard momentum?
A. It uses a smaller learning rate to ensure stability.
B. It only works for convex functions.
C. It first makes a "jump" based on previous momentum, then calculates the gradient to make a correction.
D. It calculates the gradient at the current position and then adds the momentum vector.
Answer: C. It first makes a "jump" based on previous momentum, then calculates the gradient to make a correction.

57. For what type of features is the AdaGrad algorithm noted to work particularly well?
A. Sequential data
B. Image pixels
C. Dense features
D. Sparse features
Answer: D. Sparse features.

58. What significant issue associated with AdaGrad is solved by using RMSProp?
A. Its poor performance on sparse features.
B. The diminishing learning rate issue.
C. Its inability to converge on simple problems.

D. Its high computational cost.
Answer: B. The diminishing learning rate issue.

59. The Adam optimization algorithm gets its power by combining the advantages of which two other methods? ✨
A. Standard GD and SGD
B. Nesterov Momentum and Standard Momentum
C. AdaGrad and RMSProp
D. Gradient Clipping and Batch Normalization
Answer: C. AdaGrad and RMSProp.

60. The momentum term v in the "GD with Momentum" algorithm can be interpreted as what physical concept?
A. Acceleration
B. Velocity
C. Friction
D. Mass
Answer: B. Velocity.

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A. Standard GD and SGD
B. Nesterov Momentum and Standard Momentum
C. AdaGrad and RMSProp
D. Gradient Clipping and Batch Normalization
Answer: C. AdaGrad and RMSProp.

64. A key limitation of the AdaGrad optimizer is:
a) It increases learning rate over time
b) It decreases learning rate too aggressively
c) It does not support sparse features
d) It cannot be combined with momentum

Correct Answer: b) It decreases learning rate too aggressively
Reason: AdaGrad keeps accumulating squared gradients, causing the effective learning rate to shrink over time, which may stop further learning.

65. RMSProp modifies AdaGrad by:
a) Resetting the learning rate at each epoch
b) Using an exponentially decaying average of squared gradients
c) Applying L1 regularization
d) Eliminating gradient descent entirely

Correct Answer: b) Using an exponentially decaying average of squared gradients
Reason: RMSProp controls AdaGrad's aggressive decay by applying an exponentially weighted moving average of squared gradients, stabilizing learning.

66. Which of the following optimizers combines the benefits of momentum and RMSProp?
a) SGD
b) AdaGrad
c) Adam
d) Nesterov

Correct Answer: c) Adam
Reason: Adam integrates momentum (first moment estimate) with RMSProp (second moment estimate), making it adaptive and efficient.

67. In the Adam optimizer, the two moment estimates track:
a) Bias and variance of data
b) Mean and variance of activations
c) First-order moment (mean) and second-order moment (uncentered variance) of gradients
d) Gradient clipping thresholds

Correct Answer: c) First-order moment (mean) and second-order moment (uncentered variance) of gradients
Reason: Adam maintains moving averages of gradients and squared gradients, allowing adaptive updates per parameter.

68. Which of the following best describes L1 regularization?
a) Penalizes the sum of squared weights
b) Forces weights toward sparsity by penalizing the absolute value of weights
c) Increases the learning rate
d) Reduces gradient explosion

Correct Answer: b) Forces weights toward sparsity by penalizing the absolute value of weights
Reason: L1 adds a penalty proportional to |w|, pushing some weights to exactly zero, resulting in sparse models.

69. The penalty term added in L2 regularization is proportional to:
a) |w|
b) $|w|^2$
c) 1/|w|
d) log(w)

Correct Answer: b) $|w|^2$
Reason: L2 adds squared weights to the loss, preventing very large weights and improving model generalization.

70. Which of the following is a common effect of L1 regularization?
a) Encourages large weight values
b) Produces sparse models with many zero-valued parameters
c) Increases training accuracy always
d) Removes the need for dropout

Correct Answer: b) Produces sparse models with many zero-valued parameters
Reason: L1 encourages sparsity by pushing weights to zero, which is useful for feature selection.

71. L2 regularization is particularly effective in:
a) Reducing model sparsity
b) Preventing weights from growing too large by spreading weights more evenly
c) Eliminating the bias term

d) Handling vanishing gradients

Correct Answer: b) Preventing weights from growing too large by spreading weights more evenly
Reason: L2 penalizes large weights but doesn't eliminate them, controlling model complexity without enforcing sparsity.

72. Which statement about Adam is correct?

a) Adam uses only global learning rates
b) Adam adapts the learning rate individually for each parameter
c) Adam is slower than SGD in practice
d) Adam requires no hyperparameter tuning

Correct Answer: b) Adam adapts the learning rate individually for each parameter
Reason: Adam uses both first and second moment estimates, which allow per-parameter learning rate adjustments.

73. . If a model tends to overfit, which of the following combinations is most suitable?

a) Remove all regularization and increase depth
b) Apply L1/L2 regularization along with dropout
c) Reduce dataset size
d) Train with a very high learning rate

Correct Answer: b) Apply L1/L2 regularization along with dropout
Reason: Regularization and dropout reduce model complexity and prevent over-reliance on specific neurons, improving generalization.

1. Early stopping is primarily used to:

- A. Reduce training time only
- B. Prevent overfitting by stopping when validation loss increases
- C. Improve model capacity exponentially
- D. Reduce the need for learning rate scheduling

Answer: B

2. Which criterion is most often monitored in early stopping?

- A. Training accuracy
- B. Gradient norm
- C. Validation loss/accuracy
- D. Input variance

Answer: C

3. Data augmentation helps models by:

- A. Increasing model parameters
- B. Synthesizing new, varied training samples from existing data
- C. Reducing the size of the training dataset
- D. Applying regularization penalties directly on weights

Answer: B

4. Which of the following is not a typical dataset augmentation technique?

- A. Rotation and flipping in images
- B. Adding noise to inputs
- C. Reducing the number of samples
- D. Color jittering in images

Answer: C

5. Dataset augmentation is especially important when:

- A. Data is abundant and perfectly balanced
- B. Data is scarce or imbalanced
- C. Gradient descent diverges
- D. Validation accuracy is higher than training accuracy

Answer: B

6. Parameter tying is mostly used in:

- A. Convolutional Neural Networks (CNNs)
- B. Recurrent Neural Networks (RNNs)
- C. Decision Trees
- D. k-Nearest Neighbors

Answer: B

7. The main advantage of ensemble methods is:

- A. Faster training of a single model
- B. Combining multiple models to reduce variance and improve generalization
- C. Eliminating the need for training-validation split
- D. Reducing dataset size

Answer: B

8. Dropout works by:

- A. Removing irrelevant input features permanently
- B. Randomly dropping units during training to prevent overfitting
- C. Sharing parameters across layers
- D. Scaling input features to unit variance

Answer: B

9. Batch normalization primarily helps in:

- A. Eliminating dropout
- B. Stabilizing and accelerating training by normalizing layer inputs
- C. Increasing parameter count in the model
- D. Reducing dataset variance

Answer: B

10. Which statement about batch normalization is correct?

- A. It always eliminates overfitting
- B. It adds trainable shift and scale parameters after normalization
- C. It requires the batch size to be exactly 1
- D. It decreases the number of parameters in the model

Answer: B

Q1. What is the main purpose of convolutional layers in CNNs?

- a) Reduce the dimensionality of input
- b) Extract features from input data
- c) Perform final classification
- d) Normalize pixel intensities

A: b) Extract features from input data

Q2. Which activation function is most commonly used in CNNs due to its simplicity and ability to mitigate vanishing gradients?

- a) Sigmoid
- b) Tanh
- c) ReLU
- d) Softmax

A: c) ReLU

Q3. What does the kernel (filter) do in a convolution operation?

- a) Performs matrix inversion
- b) Detects local features in the input
- c) Randomly selects pixels
- d) Reduces feature maps by pooling

A: b) Detects local features in the input

Q4. In convolution, stride = 2 means:

- a) Filter shifts by 2 pixels at a time
- b) Two filters are applied simultaneously
- c) Output will be doubled in size
- d) Padding is increased by 2

A: a) Filter shifts by 2 pixels at a time

Q5. The main motivation for using pooling layers is to:

- a) Increase the number of parameters
- b) Reduce computation and control overfitting
- c) Perform convolution in deeper layers
- d) Remove activation functions

A: b) Reduce computation and control overfitting

Q6. Which pooling method selects the strongest feature from each region?

- a) Average Pooling
- b) Max Pooling
- c) Global Average Pooling
- d) Min Pooling

A: b) Max Pooling

Q7. Which variant of convolution is mainly used for upsampling in tasks like image segmentation?

- a) Depthwise Convolution
- b) Transposed Convolution
- c) Dilated Convolution
- d) 1×1 Convolution

A: b) Transposed Convolution

Q8. Which convolution technique is widely used in MobileNet for computational efficiency?

- a) Standard Convolution
- b) Separable Convolution
- c) Dilated Convolution
- d) Global Convolution

A:b) Separable Convolution

Q9. The 1×1 convolution in CNNs is primarily used for:

- a) Detecting edges in the input
- b) Reducing or increasing the number of channels
- c) Replacing pooling operations
- d) Removing padding

A:b) Reducing or increasing the number of channels

Q10. Which of the following is a correct sequence of layers in a typical CNN?

- a) Input → Fully Connected → Pooling → Convolution → Output
- b) Input → Convolution → Activation → Pooling → Fully Connected → Output
- c) Input → Pooling → Convolution → Fully Connected → Output
- d) Input → Activation → Fully Connected → Pooling → Output

A: b) Input → Convolution → Activation → Pooling → Fully Connected → Output

1) Which domain most prominently uses CNNs for detecting tumors and lesions in images like MRIs and X■rays?
A. Financial forecasting
B. Medical imaging
C. Compiler optimization
D. Network routing
Answer: B. Medical imaging

2) In autonomous driving stacks, CNNs are primarily used for which stage?
A. Battery management
B. Perception tasks such as object and sign detection
C. Engine control
D. Map compression

Answer: B. Perception tasks such as object and sign detection

3) Which is a common CNN use in factories?
A. Compiler parsing
B. Quality inspection and defect detection
C. RF antenna tuning
D. Power grid dispatch
Answer: B. Quality inspection and defect detection

4) A hospital builds a classifier on fundus images to find hemorrhages; which strategy is routinely used to reduce overfitting in such CNN pipelines?
A. Removing validation data
B. Label smoothing
C. Heavy data augmentation
D. Zero initialization of all weights
Answer: C. Heavy data augmentation

5) Which statement about retail and e■commerce is accurate regarding CNNs?
A. CNNs cannot process product images
B. CNNs support image■based recommendations and catalog quality checks
C. CNNs only run on supercomputers
D. CNNs are unsuitable for large catalogs
Answer: B. CNNs support image■based recommendations and catalog quality checks

6) Which activation popularized by AlexNet sped up training compared to tanh/sigmoid?
A. Softsign
B. ReLU
C. GELU
D. SiLU
Answer: B. ReLU

7) Which regularization did AlexNet rely on in fully connected layers?
A. Early stopping only
B. Weight tying
C. Dropout with high keep probability
D. Dropout with drop probability around 0.5
Answer: D. Dropout with drop probability around 0.5

8) What was the role of Local Response Normalization (LRN) in AlexNet?
A. Channel■wise batch standardization
B. Encouraging lateral inhibition/competition among activations
C. Whitening inputs
D. Spatial averaging of features
Answer: B. Encouraging lateral inhibition/competition among activations

9) What technique did ZFNet introduce to interpret learned features and diagnose design issues?
A. Stochastic depth
B. Deconvolutional network visualizations
C. Squeeze■and■excitation
D. Capsule routing
Answer: B. Deconvolutional network visualizations

10) Which change best characterizes how ZFNet refined AlexNet's early layers?
A. Larger 11×11 kernels with stride 4
B. Replacing max pooling with average pooling only
C. Reducing initial receptive field and stride to mitigate aliasing
D. Removing ReLU in early layers
Answer: C. Reducing initial receptive field and stride to mitigate aliasing

1. What was the main design principle of VGG models?
a) Large convolution filters
b) Small 3x3 convolution filters used consistently
c) Randomized filter sizes
d) Fully connected layers only

✅ **Answer:** b) Small 3x3 convolution filters used consistently

2. Which activation function is consistently used in VGG-19 after each convolutional layer?
a) Sigmoid
b) Tanh
c) ReLU
d) Leaky ReLU

✅ **Answer:** c) ReLU

3. Which of the following problems does ResNet solve?
a) Overfitting in small datasets
b) Vanishing and exploding gradients in deep networks
c) Lack of feature extraction in CNNs
d) Reducing computational costs

✅ **Answer:** b) Vanishing and exploding gradients in deep networks

4. What is the key idea behind ResNet?
a) Increasing convolutional filter sizes
b) Using residual (skip) connections

c) Removing fully connected layers
d) Adding more pooling layers

✅ **Answer:** b) Using residual (skip) connections

5. What is one major drawback of VGG-19?
a) It cannot be used for transfer learning
b) It requires very high computational resources
c) It has poor accuracy compared to AlexNet
d) It cannot work with colored images

✅ **Answer:** b) It requires very high computational resources

6. Which of the following is a ResNet variant with 50 layers?
a) ResNet-18
b) ResNet-34
c) ResNet-50
d) ResNet-101

✅ **Answer:** c) ResNet-50

7. What does the computational graph of an RNN primarily represent?
a) The structure of convolutional filters
b) The flow of data through time steps
c) Only the input layer and output layer connections
d) A static representation of a feedforward network

✅ **Answer:** b) The flow of data through time steps

8. In an RNN computational graph, what is shared across all time steps?
a) Different weight matrices
b) Unique biases at each step
c) Same parameters (weights and biases)
d) Separate loss functions

✅ **Answer:** c) Same parameters (weights and biases)

9. What is the main challenge when training RNNs using computational graphs?
a) Gradient explosion and vanishing
b) High memory cost due to convolution filters
c) Lack of nonlinear activation functions
d) No sequential dependencies

✅ **Answer:** a) Gradient explosion and vanishing

10. The unrolled computational graph of an RNN is often called:
a) Feedforward network
b) Backward graph
c) Time-unfolded network
d) Convolutional chain

✅ **Answer:** c) Time-unfolded network

1. What is the function of the encoder in an encoder-decoder model?**
A) It creates the final output sequence for translation
B) It changes the input sequence into a context vector
C) It classifies input data as positive or negative
D) It adjusts model parameters
Answer: B) It changes the input sequence into a context vector[1]

2. What role does the decoder play in a sequence-to-sequence architecture?
A) It reads the original input sequence again
B) It generates each token of the output sequence using context information
C) It stores the training data
D) It finds errors in the input sequence
Answer: B) It generates each token of the output sequence using context information[1]

3. Encoder-decoder models are mostly used for which of these tasks?
A) Image recognition
B) Sequence-to-sequence tasks like translation
C) Data compression
D) Object detection
Answer: B) Sequence-to-sequence tasks like translation[1]

4. Which neural network type is usually used for working with encoder-decoder models?
A) RNN (Recurrent Neural Network)
B) CNN (Convolutional Neural Network)
C) GAN (Generative Adversarial Network)
D) SVM (Support Vector Machine)
Answer: A) RNN (Recurrent Neural Network)[1]

5. Which application best describes encoder-decoder usage?
A) Sorting words alphabetically
B) Machine translation (changing one language to another)

C) Predicting weather data
D) Detecting spam
Answer: B) Machine translation (changing one language to another)[1]

6. LSTM and GRU are better than simple RNNs because they:
A) Need less training data
B) Can remember information for longer and solve vanishing gradient problems
C) Use images instead of text
D) Are always faster

Answer: B) Can remember information for longer and solve vanishing gradient problems[1]
7. In an LSTM, which gate helps remove unneeded past information?
A) Output gate
B) Forget gate
C) Input gate
D) Reset gate
Answer: B) Forget gate[1]

8. What does a language model do in NLP?
A) It predicts the most likely next word in a sentence
B) It sorts sentences by length
C) It compresses text data
D) It translates languages without context
Answer: A) It predicts the most likely next word in a sentence[1]

9. What is special about GRU compared to LSTM?
A) It has fewer gates and a simpler structure
B) It is only used for image data
C) It cannot process sequences
D) It skips training
Answer: A) It has fewer gates and a simpler structure[1]

10. What type of data structure is best processed using Recursive Neural Networks (RvNNs)?
A) List of numbers
B) Tree-like structures
C) Random numbers
D) Tables with rows and columns

Answer: B) Tree-like structures[1]

1. Which component is present in an LSTM cell but is not explicitly present in a GRU cell?
A. Update Gate
B. Reset Gate
C. Cell State
D. Hidden State
Answer: C. Cell State

2. What is the primary challenge faced by traditional Recurrent Neural Networks (RNNs) that LSTMs and GRUs were designed to mitigate?
A. Overfitting on small datasets
B. Exploding and vanishing gradients
C. High computational cost
D. Inability to process non-sequential data
Answer: B. Exploding and vanishing gradients

3. In a Many-to-One RNN architecture, which application is given as an example in the document?
A. Language Translation
B. Image Captioning
C. Sentiment Analysis
D. Time-Series Prediction
Answer: C. Sentiment Analysis

4. What is the role of the Forget Gate in a Long Short-Term Memory (LSTM) network?
A. It controls how much new information should be added to the cell state.
B. It determines what past information should be discarded from the cell state.
C. It regulates what information should be output at the current step.
D. It calculates the potential new hidden state.
Answer: B. It determines what past information should be discarded from the cell state

5. Gated Recurrent Units (GRUs) are considered computationally more efficient than LSTMs primarily because they:
A. Use a more complex cell structure.
B. Combine the input and forget gates into a single update gate and merge the cell state and hidden state.
C. Are designed only for short-term dependencies.
D. Do not require backpropagation through time.
Answer: B. Combine the input and forget gates into a single update gate

6. Which of the following is not an application of Recurrent Neural Networks (RNNs) mentioned in
the document?
A. Time-Series Prediction
B. Speech Recognition
C. Natural Language Processing (NLP)
D. Relational Database Management
Answer: D. Relational Database Management

7. What are the two training tasks used during the pre-training of the BERT model?
A. Next Word Prediction and Sentiment Classification
B. Masked Language Modeling and Named Entity Recognition
C. Masked Language Modeling and Next Sentence Prediction
D. Question Answering and Text Summarization
Answer: C. Masked Language Modeling and Next Sentence Prediction

8. BERT is a language model based on which neural network architecture, utilizing attention mechanisms to weigh word importance?
A. Recurrent Neural Network (RNN)
B. Long Short-Term Memory (LSTM)
C. Transformer
D. Convolutional Neural Network (CNN)
Answer: C. Transformer

9. In the context of sentiment analysis using deep learning models like RNNs, what is the process of converting text into a sequence of integers called?
A. Text Translation
B. Batching
C. Text Vectorization
D. Backpropagation
Answer: C. Text Vectorization

10. What is the name of the process that expands the recurrent structure of an RNN over time steps
to illustrate how information flows and enables backpropagation?
A. Unfolding (or Unrolling)
B. Activation
C. Hidden State Calculation

D. Gating
Answer: A. Unfolding (or Unrolling)

Q1. What is the main purpose of an autoencoder?
a) Classification of images
b) Reducing dimensionality and reconstructing input
c) Making predictions on future values
d) Detecting outliers
Answer: b) Reducing dimensionality and reconstructing input

Q2. Which two main components make up an autoencoder?
a) Generator and discriminator
b) Encoder and decoder
c) Input and hidden layers
d) Regressor and classifier
Answer: b) Encoder and decoder

Q3. What does the encoder in an autoencoder do?
a) Expands the data
b) Compresses the data into a latent space
c) Filters random noise
d) Creates random samples
Answer: b) Compresses the data into a latent space

Q4. Why are autoencoders often used before classification tasks?
a) To train faster networks
b) To extract important features
c) To improve visualization only
d) To reduce overfitting always
Answer: b) To extract important features

Q5. What makes a variational autoencoder different from a basic autoencoder?
a) It uses reinforcement learning
b) It learns a probability distribution of the latent space
c) It never uses a decoder
d) It is only used for classification
Answer: b) It learns a probability distribution of the latent space

Q6. In a variational autoencoder, the latent variables are modeled as:

a) Fixed values
b) Random variables with a distribution
c) Predefined labels
d) Noise removal filters
Answer: b) Random variables with a distribution

Q7. Which trick is used in VAE to allow backpropagation through random sampling?
a) Dropout trick
b) Reparameterization trick
c) Gradient clipping
d) Batch normalization
Answer: b) Reparameterization trick

Q8. Which of the following is a common application of autoencoders?
a) Image compression
b) Word translation
c) Stock market prediction
d) Sorting data
Answer: a) Image compression

Q9. How are autoencoders used in anomaly detection?
a) They predict anomalies directly
b) They reconstruct data, and high reconstruction error indicates anomaly
c) They store anomaly patterns
d) They assign anomalies to clusters
Answer: b) They reconstruct data, and high reconstruction error indicates anomaly

Q10. Which of the following is a typical application of variational autoencoders?
a) Generating new images like faces
b) Sorting documents alphabetically
c) Performing simple arithmetic
d) Encrypting passwords
Answer: a) Generating new images like faces

Q1. Which of the following best describes a Boltzmann Machine?
a) A feedforward neural network with sigmoid activations
b) A recurrent stochastic neural network that models probability distributions
c) A decision tree-based classifier
d) A deterministic autoencoder
Answer: b

Q2. What is the energy function used in Boltzmann Machines?

Q3. Why are Restricted Boltzmann Machines (RBMs) easier to train than full BMs?
a) They use backpropagation
b) They restrict connections between hidden-hidden and visible-visible units
c) They require no biases
d) They do not use stochastic units
Answer: b

Q4. Which algorithm is commonly used to train Restricted Boltzmann Machines (RBMs)?
a) Stochastic Gradient Descent (SGD)
b) Contrastive Divergence (CD)
c) Expectation Maximization (EM)
d) Reinforcement Learning
Answer: b

Q5. What role do hidden units in an RBM play?
a) Directly provide predictions
b) Store labels for supervised learning
c) Capture dependencies and extract features from visible units
d) Act as loss functions
Answer: c

Q6. A Deep Belief Network (DBN) is primarily:
a) A stack of multiple RBMs
b) A convolutional neural network
c) A supervised deep neural network
d) A decision tree ensemble
Answer: a

Q7. What is the typical training process of a Deep Belief Network?
a) Entire network trained end-to-end with gradient descent only
b) Greedy layer-wise pre-training using RBMs, followed by fine-tuning
c) Random weight initialization without pre-training
d) Supervised learning only on visible units
Answer: b

Q8. Which of the following applications has used RBMs successfully?
a) Image classification
b) Netflix recommendation system
c) Text-to-speech synthesis

d) Reinforcement learning for robotics
Answer: b

Q9. Why are DBNs considered important in deep learning history?
a) They were the first models to outperform CNNs on ImageNet
b) They provided a way to effectively pre-train deep networks before large datasets were available
c) They eliminated the vanishing gradient problem entirely
d) They replaced LSTMs in sequence modeling
Answer: b

Q10. In a DBN, after pre-training with RBMs, the network is usually:
a) Used directly without further modification
b) Fine-tuned with supervised backpropagation
c) Converted into a decision tree
d) Used only for dimensionality reduction
Answer: b

Q1. Which website is widely used for finding open-source datasets in machine learning and deep learning?
a) GeeksforGeeks
b) Kaggle
c) Facebook
d) Stack Overflow
Answer: b) Kaggle

Q2. Which of the following is a common file format for tabular datasets?
a) .csv
b) .jpg
c) .wav
d) .mp4
Answer: a) .csv

Q3. Image datasets are usually stored using which of the following file types?
a) .xls
b) .csv
c) .jpg/.png
d) .txt
Answer: c) .jpg/.png

Q4. What's a common drawback of label encoding for nominal categories?
a) Introduces false ordinal relationships
b) Too slow
c) Increases dimensionality
d) Requires normalization
Answer: a) Introduces false ordinal relationships

Q5. What is the purpose of an activation function in a neural network?
a) It introduces non-linearity to the network
b) It determines the output of a neuron
c) It helps in backpropagation
d) All of the above
Answer: d) All of the above

Q6. Which of the following is not a popular deep learning framework?
a) TensorFlow
b) PyTorch
c) Keras
d) Scikit-learn
Answer: d) Scikit-learn

Q7. What is the role of gradient descent in deep learning?
a) To minimize the loss function
b) To find the optimal weights for the neural network
c) To update the network's parameters
d) All of the above
Answer: d) All of the above

Q8. What is the vanishing gradient problem?
a) When the gradients in a deep neural network become extremely small
b) When the gradients in a deep neural network become extremely large
c) When the output does not change with input
d) None of these
Answer: a) When the gradients in a deep neural network become extremely small

Q9. Which layer extracts spatial features from input images in CNNs?
a) Fully connected layer
b) Convolutional layer

c) Pooling layer
d) Dropout layer
Answer: b) Convolutional layer

Q10. How many layers are there in deep learning algorithms?
a) One
b) Two
c) More than two
d) Ten
Answer: c) More than two

1. Which of the following is the official high-level API integrated into TensorFlow?
a) PyTorch
b) Keras
c) MXNet
d) NumPy
Answer: b) Keras

2. TensorFlow is developed and maintained by:
a) Facebook AI Research
b) Google
c) Microsoft
d) Apache Foundation
Answer: b) Google

3. Which deep learning framework is maintained by Facebook's AI Research lab (FAIR)?
a) TensorFlow
b) Keras
c) PyTorch
d) Theano
Answer: c) PyTorch

4. Which Python library is mainly used for data manipulation and analysis, especially with time series and numerical tables?
a) NumPy
b) Pandas
c) Seaborn
d) Matplotlib
Answer: b) Pandas

5. NumPy is mainly useful for:
a) Data visualization
b) Creating neural networks
c) Processing multidimensional arrays and matrices
d) Clustering datasets
Answer: c) Processing multidimensional arrays and matrices

6. Which Python library is widely used for creating plots such as line plots, scatter plots, histograms, and heatmaps?
a) Pandas
b) Matplotlib
c) Seaborn
d) scikit-learn
Answer: b) Matplotlib

7. Seaborn is built on top of which Python library?
a) TensorFlow
b) PyTorch
c) Matplotlib
d) NumPy
Answer: c) Matplotlib

8. Which Python library provides tools for classification, regression, clustering, and feature selection?
a) Pandas
b) Seaborn
c) scikit-learn
d) Theano
Answer: c) scikit-learn

9. MXNet is especially known for its:
a) Symbolic and imperative programming support
b) Ability to only work with Python
c) Exclusive use in visualization
d) Lack of scalability
Answer: a) Symbolic and imperative programming support

10. Theano was primarily created for:
a) Data visualization
b) Numerical computing and machine learning

c) Clustering and regression tasks
d) Building dashboards
Answer: b) Numerical computing and machine learning

Program 1: Linear Regression
Q1. In the linear regression program, which library function is used to calculate the mean squared
error?
a) sklearn.model_selection.train_test_split
b) sklearn.metrics.mean_squared_error
c) numpy.mean
d) sklearn.linear_model.LinearRegression
Answer: b) sklearn.metrics.mean_squared_error
Q2. What is the purpose of reshaping X_train and X_test into (-1,1) in the linear regression program?
a) To normalize the values
b) To prepare 1D arrays for plotting
c) To convert the data into a 2D array suitable for scikit-learn models
d) To avoid overfitting
Answer: c) To convert the data into a 2D array suitable for scikit-learn models

Program 2: Titanic Data Preprocessing
Q3. Which method is used in the Titanic program to replace missing values in the Age column?
a) Mean imputation
b) Median imputation
c) Mode imputation
d) Zero imputation
Answer: b) Median imputation
Q4. In the Titanic dataset preprocessing, which encoding technique is applied to the Embarked column?
a) Label Encoding only
b) Binary Encoding
c) One-Hot Encoding
d) Frequency Encoding
Answer: c) One-Hot Encoding
Q5. Which algorithm is used in the program to detect outliers with contamination = 0.05?
a) DBSCAN
b) Isolation Forest
c) Z-Score
d) IQR

Answer: b) Isolation Forest

Program 3: Image Enhancement

Q6. Which OpenCV function is used to perform histogram equalization in the image enhancement
program?
a) cv2.equalizeHist()
b) cv2.equalizeImage()
c) cv2.equalizeContrast()
d) cv2.histEqual()
Answer: a) cv2.equalizeHist()
Q7. What type of morphological operation is applied in the program using cv2.morphologyEx()?
a) Dilation
b) Erosion
c) Opening
d) Closing
Answer: c) Opening
Q8. Which function is used to binarize the grayscale image in the image enhancement program?
a) cv2.adaptiveThreshold()
b) cv2.threshold()
c) cv2.Canny()
d) cv2.binarize()
Answer: b) cv2.threshold()

Program 4: Feedforward Neural Network (CIFAR-10)
Q9. How many images are present in the CIFAR-10 training set according to the program description?
a) 10,000
b) 60,000
c) 50,000
d) 5,000
Answer: c) 50,000
Q10. In the feedforward neural network experiment, which step is performed before training the model?
a) Dropout regularization
b) Data normalization and one-hot encoding of labels
c) Batch normalization
d) PCA dimensionality reduction
Answer: b) Data normalization and one-hot encoding of labels

MCQs on Deep Learning Lab Topics

Q1. What is the input image size in the MNIST dataset?

- a) 32×32
- b) 28×28
- c) 64×64
- d) 224×224

Answer: b) 28×28

Q2. In CNN architectures like LeNet, which layer typically reduces spatial dimensions?

- a) Dense layer
- b) Convolutional layer
- c) Pooling layer
- d) Flatten layer

Answer: c) Pooling layer

Q3. What activation function is most commonly used in convolutional layers?

- a) Sigmoid
- b) Tanh
- c) ReLU
- d) Softmax

Answer: c) ReLU

Q4. What is the final activation function for MNIST digit classification (10 classes)?

- a) ReLU
- b) Sigmoid
- c) Softmax
- d) Tanh

Answer: c) Softmax

Q5. Xavier initialization is particularly useful for:

- a) Layers with ReLU activation
- b) Layers with Sigmoid/Tanh activation
- c) Pooling layers
- d) Output layers only

Answer: b) Layers with Sigmoid/Tanh activation

Q6. Kaiming initialization is designed to work best with:

- a) ReLU activation
- b) Tanh activation
- c) Softmax activation
- d) Linear activation

Answer: a) ReLU activation

Q7. What is the main purpose of dropout in neural networks?

- a) To increase model parameters
- b) To prevent overfitting
- c) To speed up training
- d) To improve gradient flow

Answer: b) To prevent overfitting

Q8. Which optimizer adapts the learning rate for each parameter individually?

- a) SGD
- b) RMSProp
- c) Adam
- d) Momentum

Answer: c) Adam

Q9. Why do MNIST images need resizing before being fed into VGG-19?

- a) VGG-19 requires 32×32 images
- b) VGG-19 requires 224×224 RGB images
- c) VGG-19 requires grayscale images
- d) VGG-19 requires flattened 784-dimension vectors

Answer: b) VGG-19 requires 224×224 RGB images

Q10. Which layers of a pre-trained model like VGG-19 are usually replaced when applying transfer learning?

- a) Convolutional layers
- b) Pooling layers
- c) Fully connected classification layers
- d) Batch normalization layers

Answer: c) Fully connected classification layers

Which type of RNN layer is used in the model implemented in the code?
A) LSTM
B) GRU
C) SimpleRNN
D) Bidirectional RNN

What is the activation function used in the final Dense layer of the RNN model?
A) ReLU
B) Sigmoid
C) Tanh
D) Softmax

Why is it necessary to ensure that all input sequences (movie reviews) have the same length before feeding them into an RNN model?
A) Because RNNs can only process fixed-length inputs at each timestep
B) Because variable-length sequences reduce the size of the vocabulary
C) Because the model cannot handle unknown words
D) Because longer reviews always contain more sentiment information

In the IMDB dataset preprocessing step, why do we restrict the vocabulary to the top 10,000 most frequent words instead of using all unique words?
A) To reduce computational complexity and prevent the model from overfitting on rare words
B) To remove all stop words and punctuation
C) To ensure that every review has exactly 10,000 words
D) To discard all positive reviews beyond the 10,000th word

1. In the context of sentiment analysis on text data, sequences are often padded to a fixed length before being fed into a recurrent neural network (RNN). What is the primary purpose of

this padding step?
A) To increase the vocabulary size by adding special tokens.
B)To ensure that all input sequences have the same dimensions, which is a requirement for batch processing.
C)To improve the model's accuracy by providing more contextual information.
D)To reduce the computational cost by shortening all sequences.

2. When comparing Long Short-Term Memory (LSTM) and Gated Recurrent Unit (GRU) architectures, a key theoretical difference is that:
A) LSTMs have a forget gate, while GRUs combine the input and forget gates into a single update gate.
B)GRUs are fundamentally incapable of handling long-term dependencies, whereas LSTMs are designed for them.
C)LSTMs use significantly fewer parameters than GRUs, making them less computationally complex.
D)GRUs always outperform LSTMs on sequential data tasks due to their simplified gating Mechanism.

3. The Embedding layer at the start of a model for text classification serves which primary function?
A) It performs part-of-speech tagging to identify grammatical structure.
B)It converts integer-encoded words into dense vector representations where semantic meaning is captured by spatial relationships. C)It acts as a final classification layer, outputting a probability for each class.
D)It removes stop words and punctuation from the input sequences.

4. A model is trained for sentiment analysis and achieves high accuracy on the training data but significantly lower accuracy on the validation and test data. This phenomenon is most indicative of:
A) Underfitting, where the model has failed to learn the underlying patterns in the training data.
B)Optimal convergence, indicating the model has learned the task perfectly.
C)Overfitting, where the model has learned the training data too well, including its noise, and fails to generalize to unseen data.
D)A low learning rate, causing the model to learn too slowly.

1.In time series forecasting, a MinMaxScaler is often applied to the data before training a model. What is the primary theoretical reason for this normalization step when using a SimpleRNN or similar neural network?
a) To convert the time series data from a cyclical format into a linear trend.

b) To reduce the impact of outliers and ensure all features contribute equally to the model's learning process, aiding in stable and faster convergence.
c) To introduce a random noise component that helps prevent the model from overfitting.
d) To segment the data into distinct seasonal components for independent analysis.

2.The concept of using past values (e.g., n_input = 3) to predict the next value in a sequence is fundamental to time series forecasting. What is thi
a) Dimensionality Reduction
b) Creating a Rolling Window or Look-back Period
c) Seasonal Decomposition
d) Feature Embedding

3.A SimpleRNN (Vanilla RNN) with a tanh activation function is used in the hidden layer. What is
a well-known theoretical limitation of this architecture when dealing with long sequences?
a) It is computationally prohibitively expensive compared to other models.
b) It suffers from the vanishing/exploding gradient problem, making it difficult to learn long-range
dependencies.
c) It can only model linear relationships and fails to capture any non-linearity in the data.
d) It requires the input sequence to be of a fixed, short length, unlike more advanced RNNs.

4.The model's final layer uses a linear activation function (implied by the absence of a non-linear activation like 'sigmoid' or 'softmax' on the single Dense unit). What does this choice indicate about the nature of the forecasting task?
a) The task is a multi-class classification problem, predicting a category for each time step.
b) The task is a binary classification problem, predicting if the stock price will go up or down.
c) The task is a regression problem, predicting a continuous numerical value (like a stock price).
d) The task involves reconstructing the input sequence, making it an unsupervised learning problem.

Q1. Which of the following is the default programming language supported in Jupyter Notebook?

- a) Java
- b) Python
- c) C++
- d) R

✅ Answer: b) Python

Q2. What does TensorFlow primarily provide?

- a) Word processing tools
- b) Deep learning and numerical computation library
- c) Operating system environment
- d) Database management system

✅ Answer: b) Deep learning and numerical computation library

Q3. In a shallow encoder–decoder network, what is the main role of the encoder?

- a) To generate random noise for training
- b) To compress the input sequence into a fixed-length context vector
- c) To expand the input vocabulary
- d) To normalize the dataset

✅ Answer: b) To compress the input sequence into a fixed-length context vector

Q4. What is the main advantage of using GPU in deep learning?

- a) Better accuracy
- b) Faster computation for matrix operations
- c) More storage capacity
- d) Larger vocabulary size

✅ Answer: b) Faster computation for matrix operations

Q5. Which type of neural network is commonly used in shallow encoder–decoder models for machine translation?

- a) Convolutional Neural Network (CNN)
- b) Recurrent Neural Network (RNN) with LSTM/GRU
- c) Generative Adversarial Network (GAN)
- d) Autoencoder without decoder

✅ Answer: b) Recurrent Neural Network (RNN) with LSTM/GRU

Q6. In machine translation tasks, why is padding required for sequences?

- a) To reduce vocabulary size
- b) To make sequences of equal length

- c) To improve model accuracy
- d) To convert text into vectors

✅ Answer: b) To make sequences of equal length

Q7. Which of the following is an open-source platform for interactive computing widely used in data science?

- a) MySQL
- b) Jupyter Notebook
- c) TensorFlow Hub
- d) Windows PowerShell

✅ Answer: b) Jupyter Notebook

Q8. In a shallow encoder–decoder architecture, what is the purpose of the decoder?

- a) To generate output sequences from the encoder's context vector
- b) To tokenize input text into numbers
- c) To pad input sequences
- d) To reduce the vocabulary size

✅ Answer: a) To generate output sequences from the encoder's context vector

Q9. Which component of deep learning is responsible for adjusting weights during training?

- a) Activation function
- b) Optimizer
- c) Dropout
- d) Loss function

✅ Answer: b) Optimizer

Q10. In TensorFlow/Keras, which method is used to start model training?

- a) model.evaluate()
- b) model.fit()
- c) model.predict()
- d) model.compile()

✅ Answer: b) model.fit()

Q1. How many layers are there in deep learning algorithms?
a) One
b) Two
c) More than two
d) Ten
Answer: c) More than two

Q2. The first layer in a neural network is called the:
a) Output layer
b) Input layer
c) Hidden layer
d) Bias layer
Answer: b) Input layer

Q3. Which neural network has only one hidden layer between input and output?
a) Shallow neural network
b) Deep neural network
c) Convolutional neural network
d) Recurrent neural network
Answer: a) Shallow neural network

Q4. Which activation function is commonly used in hidden layers?
a) Sigmoid
b) ReLU
c) Tanh
d) Softmax
Answer: b) ReLU

Q5. What is the main purpose of data preprocessing in deep learning?
a) To increase dataset size
b) To prepare and clean the raw data before training
c) To build neural network architecture
d) To encode labels
Answer: b) To prepare and clean the raw data before training

Q6. Data augmentation is used for:
a) Increasing training data volume by transformations
b) Normalizing data

c) Encoding categorical variables
d) Reducing network layers
Answer: a) Increasing training data volume by transformations

Q7. An example of a tabular data format is:
a) .csv
b) .jpeg
c) .wav
d) .mp4
Answer: a) .csv

Q8. What does 'epoch' represent?
a) One complete pass through the training data
b) Number of features in dataset
c) Size of a batch
d) Number of layers
Answer: a) One complete pass through the training data

Q9. What is batch size in deep learning training?
a) Number of training epochs
b) Number of samples processed before model update
c) Number of neurons in a layer
d) Number of layers in a network
Answer: b) Number of samples processed before model update

Q10. What is the role of the validation set?
a) To train the model
b) To tune model hyperparameters
c) To make final predictions
d) To preprocess data
Answer: b) To tune model hyperparameters